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Heterogeneous Markups Cyclicalities and Monetary Policy

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ABSTRACT

This paper studies aggregate markup cyclicalities through firm-level heterogeneity, reallocation, and aggregation. Using the U.S. firm data for 1990–2016, we find that markups are procyclical for young firms and countercyclical for older firms following monetary policy shocks. Economic activity also reallocates modestly toward young firms. As the firm population ages, the aggregate markup response shifts from acyclical or mildly procyclical to countercyclical. These findings help reconcile conflicting evidence on markup cyclicalities and imply that firm demographics shape monetary-policy transmission and the roles of demand and supply shocks in business cycles.

JEL Classification: E32, E52, D22, L11

1 | Introduction

The cyclicalities of markups conditional on economic shocks plays a central role in sticky-price New Keynesian models and in the transmission of monetary policy (e.g., Galí 2015). In this framework, countercyclical markups reflect firms' incentives to reduce prices during downturns and increase them during booms; this behavior moderates inflation in recessions and amplifies it in expansions, aligning the cyclicalities of inflation with that of GDP. Yet, despite the substantial attention received by researchers, consensus on markup cyclicalities remains elusive, with leading studies reaching opposite conclusions (e.g., Bils et al. 2018; Nekarda and Ramey 2020).

A key challenge lies in the fact that this literature has historically relied on aggregate markup proxies, reflecting both data limitations and the prevalence of representative-agent frameworks. However, price-setting decisions — and hence markup adjustments — occur at the firm level (e.g., Mongey 2021). We therefore revisit the cyclicalities of the aggregate markup, focusing on its response to monetary policy (MP) shocks, using state-of-the-art

industrial-organization (IO) measures of firm-level markups and a micro-to-macro strategy that explicitly accounts for firm-level heterogeneity, reallocation, and aggregation.

Our approach yields three insights. First, in the United States, markups of old firms are countercyclical to MP shocks, whereas those of young firms are procyclical. Second, aggregating these heterogeneous firm-level responses implies an aggregate markup that is countercyclical over our sample. Third, changes in the firm age distribution have shifted aggregate markup cyclicalities over time — from acyclical or mildly procyclical to countercyclical — potentially offering a unified explanation for previous empirical findings.

We begin by reviewing the conventional New Keynesian (NK) logic linking demand shocks, markups, and inflation. We then connect this aggregate view to firm-level markup responses and develop a micro-to-macro framework separating three forces: firm-level markup cyclicalities, the reallocation of economic activity across firms, and aggregation. In so doing, we consider three dimensions of firm heterogeneity that the literature suggests

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could directly inform the source and direction of firm-level markup cyclicity: financial frictions, demand frictions, and nominal pricing frictions.

We implement our strategy using quarterly Compustat data. Although Compustat covers only publicly listed firms, it offers high-frequency information on firm financial statements and input expenditures, reports firm age and industry classification, and includes firms that account for much of U.S. business-cycle fluctuations (Crouzet and Mehrotra 2020). We measure firm-level markups using state-of-the-art IO methods, as in De Loecker et al. (2020),¹ and exploit high-frequency identified MP shocks from Jarociński and Karadi (2020) as our measure of demand shocks.

To estimate firm-level markup responses, we use local projections following Jordà (2005). We interact MP shocks with firm characteristics that proxy for different sources of heterogeneity: firm age (Cloyne et al. 2023), assets (Gertler and Gilchrist 1994), leverage (Ottonello and Winberry 2020), and liquidity (Jeenas 2024) for financial frictions; sales share and markups for demand frictions; and sector-time fixed effects to absorb sectoral differences in nominal pricing frictions.

Our firm-level results show that age is the strongest predictor of heterogeneous markup responses to MP shocks. Old firms increase markups after a contractionary MP shock, consistent with a countercyclical behavior, while young firms lower their markups, consistent with procyclicality. We also find a small reallocation of economic activity from old to young firms after MP shocks, in line with the theoretical mechanism in Baqaee et al. (2024). The age gradient is consistent with both financial and demand-side mechanisms: young firms may be more financially constrained and therefore reduce markups to generate cash flows, while older firms may face more inelastic demand because they are more established in their markets.

Following our micro-to-macro approach, we next aggregate the firm-level responses using variable-cost weights, which are theoretically grounded (e.g., Grassi 2017; Edmond et al. 2023) and make our aggregate markup comparable to the inverse labor share proxy commonly used in the literature. Although estimates are not always strongly significant, the aggregate markup increases after a contractionary MP shock over our sample, consistent with the predictions of sticky-price NK models.

We then show that changes in the age distribution of firms are central for understanding how aggregate markup cyclicity has evolved over time. A growing literature documents a sustained decline in firm formation and in the exit rate of older firms, leading to a progressive aging of the U.S. firm population (e.g., Decker et al. 2016; Hathaway and Litan 2014). In our framework, this shift matters because older firms have countercyclical markups and account for a growing share of economic activity. Assessing the impact of firm aging on aggregate markup cyclicity over time, we find that early in our sample – when young, procyclical firms were more prevalent – the aggregate markup was acyclical or mildly procyclical. In the latter part of the sample, as older, countercyclical firms become dominant, the aggregate markup turns countercyclical, more strongly so than in estimates based on the full sample. We corroborate these results by directly estimating the response

of the aggregate markup to MP shocks over time, which yields consistent conclusions.

This result suggests firm aging affects business-cycle fluctuations beyond its impact on factor shares and long-run trends (e.g., Hopenhayn et al. 2022; Karahan et al. 2024) and helps reconcile conflicting evidence in the aggregate markup literature. Studies using longer samples that largely predate the aging of the firm population tend to find mildly procyclical markups (e.g., Cantore et al. 2021; Nekarda and Ramey 2020), whereas studies focusing on more recent periods identify a role for countercyclical markups (e.g., Bils et al. 2018). Accounting for firm-level heterogeneity and aggregation, our approach shows how changes in the firm distribution can generate these divergent findings.

Finally, we discuss two additional macroeconomic implications of our findings. First, using a closed-form comparative static, we demonstrate that the evolution of aggregate markup cyclicity could materially affect the transmission of monetary policy. Second, we show that the countercyclical response of the aggregate markup conditional on MP shocks – contrasted with its unconditional procyclical behavior – points to an important role for supply shocks in driving business cycle fluctuations.

Literature Review. We contribute to the empirical literature on the cyclicity of aggregate markups. Studies using aggregate data have reached differing conclusions, including Bils (1987), Bils and Kahn (2000), Cantore et al. (2021), Galeotti and Schiantarelli (1998), Gali et al. (2007), Hall (2014), Bils et al. (2018), Nekarda and Ramey (2020), and Rotemberg and Woodford (1999). Note that these works rely on aggregate markup proxies and typically treat aggregation as innocuous, abstracting from firm-level heterogeneity, across-firm reallocation, and changes in the firm distribution that may shape the aggregate response.

A separate strand studies the unconditional cyclicity of firm-level markups, including Afrouzi and Caloi (2022), Alati (2020), Burstein et al. (2020), and Hong (2017). While informative about firm-level pricing, unconditional evidence is difficult to map into theories whose predictions depend on the shock considered: sticky-price New Keynesian models, for example, predict countercyclical markups conditional on demand shocks but procyclical markups conditional on productivity shocks. Closely related to our analysis, Meier and Reinelt (2022) document an increase in markup dispersion following MP shocks, while Santos et al. (2022) study the response of average firm-level markups to demand and productivity shocks.

We bridge these strands by providing evidence on *heterogeneous* firm-level markup cyclicity *conditional* on MP shocks, and by tracing how aggregation shapes the resulting aggregate markup response. Accounting jointly for firm-level heterogeneity, reallocation, and changes in the firm age distribution offers a unified explanation for the contrasting findings in the aggregate markup literature across sample periods.

Our paper also contributes to the literature on heterogeneous firm-level effects of monetary policy. Existing work studies the heterogeneous impact of MP shocks on firm-level investment (Gertler and Gilchrist 1994; Crouzet and Mehrotra 2020; Ottonello and Winberry 2020; Deng and Fang 2022; Cloyne et al. 2023;

Anderson and Cesa-Bianchi 2024; González et al. 2024; Jeenas 2024; Jungherr et al. 2024), stock-prices (Ippolito et al. 2018; Darmouni et al. 2022; Gürkaynak et al. 2022), and debt maturity (Fabiani et al. 2020). We add to this literature by providing, to the best of our knowledge, the first evidence of heterogeneity in firm-level markup responses to MP shocks — a key mechanism in sticky-price New Keynesian models. We further show that firm age shapes this heterogeneity and, through aggregation, the cyclicity of the aggregate markup.

Outline. Section 2 presents the conceptual framework used throughout the paper. Section 3 details our data and measures. Section 4 outlines our empirical strategy and firm-level results. Section 5 discusses the aggregate results. Section 6 concludes.

2 | Conceptual Framework

This section frames our empirical analysis. We first review why aggregate markup cyclicity is central in standard NK models; we then link aggregate markup dynamics to firm-level markup responses; finally, we discuss distinct theories that could ground the macroeconomic role of the heterogeneous cyclicity of firm-level markups.

Theoretical Role of Markups in NK Models. The markup of price over marginal cost is central to sticky-price New Keynesian macroeconomic models (Galí 2015). To see this, consider the following firm-level NK Phillips Curve (hereafter NKPC):

$$\pi_{i,t} = \beta \mathbb{E}_t [\pi_{i,t+1}] - \lambda_i \Delta_0 \log \mu_{i,t} = -\lambda_i \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t [\Delta_h \log \mu_{i,t+h}], \quad (1)$$

where $\pi_{i,t}$ is firm-level inflation, β is the discount factor, λ_i is the slope of the firm-level NKPC, $\mu_{i,t}$ is the firm-level markup, and $\Delta_h \log \mu_{i,t+h} \equiv \log \mu_{i,t+h} - \log \mu_i$ denotes the cumulative log deviation of the markup from its steady-state value at horizon h .

Equation (1) posits that current inflation is negatively related to current and expected future deviations of markups from their desired levels. Intuitively, if firms expect future markups to be below their desired level, sticky prices induce them to raise prices already today. This logic is what links markups and inflation in the NKPC.

In the standard NK model, demand shocks — including MP shocks — move output and firms' marginal costs in the same direction. When prices cannot adjust freely, markups over marginal costs instead move in the opposite direction. Thus, a positive demand shock raises output, lowers markups below desired levels, and generates higher inflation through the NKPC. This mechanism delivers the expected co-movement between output and inflation, and Debortoli and Galí (2022); (2024) show that the same logic also operates in two-agent and heterogeneous-agent NK models.

Micro-to-Macro Link. While much of the empirical literature has focused on aggregate markups due to data constraints, most price-setting decisions — and therefore markup movements — are likely to occur at the firm level. Here, we present a conceptual framework to relate firm-level markups to aggregate markups

(and, later, in Section 5.3.1 to inflation) empirically. We start by defining the aggregate markup $\mathcal{M}$ as:

$$\mathcal{M}_t = \sum_i \omega_{i,t}^v \mu_{i,t}, \quad (2)$$

where $\omega_{i,t}^v$ is the firm-level variable-cost weight, and $\mu_{i,t}$ is the firm-level markup. We use this cost-weighted measure because, in a broad class of models including the NK framework, it is the theoretically welfare-relevant object capturing the aggregate wedge between prices and marginal costs (Grassi 2017; Edmond et al. 2023). Moreover, under standard assumptions such as labor being the primary variable input, Equation (2) aligns with the inverse labor share, the standard aggregate markup proxy in the cyclicity literature (e.g., Nekarda and Ramey 2020).

Log-linearizing Equation (2) around the steady state, the impulse response function (IRF) of the aggregate markup to a demand shock at horizon h is written as:

$$\frac{\partial \Delta_h \log \mathcal{M}_{t+h}}{\partial \varepsilon_t^m} = \underbrace{\sum_i \frac{\omega_i^v \mu_i}{\mathcal{M}} \frac{\partial \Delta_h \log \mu_{i,t+h}}{\partial \varepsilon_t^m}}_{\text{Firm-level effect}} + \underbrace{\sum_i \frac{\mu_i}{\mathcal{M}} \frac{\partial \Delta_h \omega_{i,t+h}^v}{\partial \varepsilon_t^m}}_{\text{Reallocation effect}}, \quad (3)$$

with $\sum_i \partial \Delta_h \omega_{i,t+h}^v / \partial \varepsilon_t^m = 0$, where ε_t^m is the demand shock, $\partial \Delta_h \log \mu_{i,t+h} / \partial \varepsilon_t^m$ is the firm-level markup IRF, and $\partial \Delta_h \omega_{i,t+h}^v / \partial \varepsilon_t^m$ is the firm-level IRF of variable-cost weights. The first term in Equation (3) captures how demand shocks affect markups *within* firms; the second captures how demand shocks reallocate costs *across* firms.² Equation (3) will guide our subsequent empirical analysis by separating the role of heterogeneous firm-level markup responses from the role of cost-reallocation after demand shocks.

Dimensions of Firm-Level Heterogeneity in Markup Responses to Shocks. Theoretical studies have identified three main forces that can generate heterogeneous markup responses across firms: financial frictions, demand frictions, and pricing frictions.

Financial frictions. Recent work emphasizes that financial frictions shape firms' pricing decisions over the business cycle (e.g., Gilchrist et al. 2017; Kim 2021; Meinen and Soares 2022). Since direct measures of financial constraints are difficult to implement with available firm data, we use standard proxies: firm age (Cloyne et al. 2023), assets (Gertler and Gilchrist 1994), leverage (Ottonello and Winberry 2020), and liquidity (Jeenas 2024).

Demand frictions. Heterogeneous markup responses to aggregate shocks may also arise from oligopolistic-like forces (e.g., Gopinath and Itskhoki 2010; Klenow and Willis 2016; Mongey 2021; Wang and Werning 2022; Baqaee et al. 2024; Alvarez et al. 2022), customer capital (e.g., Ravn et al. 2006; Nakamura and Steinsson 2011; Argente and Yeh 2022), or accumulated intangibles (e.g., Bronnenberg et al. 2022). These strands of literature suggest that relative firm size may be central for differential markup responses. We proxy relative size using firms' sales shares within narrowly defined industries and their markups, both of which closely relate to relative size in this class of models.

Pricing frictions. Finally, Meier and Reinelt (2022) theoretically show that heterogeneous nominal rigidities can generate heterogeneous markup responses to shocks. Since, to our knowledge, available measures of nominal rigidities are only sectoral, our empirical analysis controls for this source of variation with sector-time fixed effects, identifying firm-level heterogeneity net of sectoral differences in price rigidity.

3 | Data Sources

Here, we outline the data used for the analysis, detailing our firm-level sources and the series of MP shocks. Following that, we describe our measure of firm-level markups.

3.1 | Firm-Level Variables and Monetary Policy Shocks

Firm-level data. We use quarterly Compustat data, which provide balance-sheet information for North American publicly listed firms, and have three advantages for our purposes. First, its quarterly frequency is well suited to business-cycle analysis. Second, Compustat contains information on firm age, sales, input expenditures, capital, liabilities, and industry classification. Third, while publicly listed firms are only a subset of all the U.S. firms, they are large and account for a sizable share of aggregate activity, including approximately 30% of employment (Davis et al. 2006) and much of business-cycle fluctuations (e.g., Crouzet and Mehrotra 2020).³ Our sample runs from 1990q1 to 2016q4, matching the availability of the MP shock series described below, and Online Appendix A.1 details its construction.

We measure firm-level production using sales (SALEQ), variable inputs using cost of goods sold (COGSQ), tangible capital using gross property, plant, and equipment (PPEQTQ), overhead costs using selling, general, and administrative expenses (XSGAQ), and assets using total assets (ATQ). Liquidity is cash and short-term investments over assets, CHEQ/ATQ, while leverage is short (DLCQ/ATQ) plus long-term liabilities over assets (DLTTQ/ATQ). Finally, relative firm size is measured as a firm's sales share among firms in the same 4-digit NAICS sector. Online Appendix A.1 reports summary statistics.

Two limitations are worth noting. First, Compustat measures corporate age, that is, time since incorporation into the dataset, rather than true firm age. We therefore validate our findings using Jay Ritter's database, which contains founding years for a subset of Compustat firms. Second, Compustat does not provide firm-level deflators, so our baseline measure is based on revenues rather than physical quantities. We discuss the implications of this limitation for the measurement of markups in Section 3.2.

MP Shock Series. We use MP shocks as identified demand shocks. Our benchmark series is from Jarociński and Karadi (2020), who refine the standard identification of high-frequency MP shocks by accounting for the information revealed to private agents by Fed announcements. The series is constructed from interest-rate surprises, measured as changes in Fed Funds Futures within 30-min windows around policy announcements, and has been widely used in empirical work. For robustness, we

also use the alternative MP shock series from Gürkaynak et al. (2005).

Additional Data Sources. After merging Compustat with the MP shock series, we add quarterly macroeconomic controls: GDP growth, Consumer Price Index (CPI) growth, the Excess Bond Premium (EBP), and the 1-Year Treasury rate change, all from FRED.

3.2 | Markups Measurement

Firm-level markups are our main object of interest. Since prices and marginal costs are not directly observed, we follow the production-function approach of De Loecker and Warzynski (2012), as implemented by De Loecker et al. (2020). Under cost minimization for given prices and demand, the markup of firm i is written as:

$$\mu_{i,t} = \mathcal{E}_{s,t}^v \frac{P_{i,t} Q_{i,t}}{P_{i,t}^v X_{i,t}^v}, \quad (4)$$

where $\mathcal{E}_{s,t}^v$ is the elasticity of output with respect to the variable input X^v , $P_{i,t} Q_{i,t}$ is sales, and $P_{i,t}^v X_{i,t}^v$ is variable input expenditure. Thus, measuring markups requires sales, variable input expenditure, and an output elasticity. In our baseline, sales and variable input expenditure are measured using SALEQ and COGSQ, while output elasticities are the sector-year estimates from De Loecker et al. (2020). Online Appendix A.1 reports summary statistics for the resulting markup measure. In Online Appendix A.2.7, we verify the robustness of our results using elasticities estimated at the quarter-sector level under Cobb-Douglas and Translog production functions.

Several well-known concerns apply to this popular estimator. To begin, measurement error can add noise to firm-level markups. Moreover, because Compustat contains revenues rather than physical quantities, the use of revenue elasticities in Equation (4) may differ from the output elasticities required by theory; Bond et al. (2021) also raise concerns about the first-stage inversion used in standard IO production-function estimators. Finally, Flynn et al. (2019) and Gandhi et al. (2020) show that estimating gross-output production functions can complicate the identification of variable-input elasticities.

Our empirical strategy addresses these concerns in three ways. First, our baseline specifications include firm and sector-time fixed effects, which absorb permanent measurement differences across firms, time-varying sectoral shocks, and much of the variation associated with output elasticities. As a result, identification relies primarily on changes in the ratio of sales to variable input expenditure. Second, our object of interest is the *change* in firm-level markups after aggregate shocks, not the level of markups per se. This distinction is relevant because De Ridder et al. (2026) show that markup changes computed using revenue elasticities and output-based elasticities are highly correlated. Third, we check robustness using alternative markup measures. In particular, Online Appendix A.2.8 uses the Lerner index employed by Baqaee and Farhi (2020),⁴ while additional exercises compute markups using output elasticities from dynamic panel estimators

with restricted and unrestricted returns to scale, which avoid the first-stage inversion required by standard IO approaches.

A final caveat is that Compustat only allows us to measure the variable input bill with COGSQ, which combines labor and material costs. If firms have monopsony power, our markup measure may partly capture that margin as well (e.g., Yeh et al. 2022). While we follow the literature in using this estimator as our baseline markup measure, this caveat should be kept in mind when interpreting the results.

4 | Firm-Level Analysis

This section studies whether firm characteristics predict heterogeneous markup responses to MP shocks. We first describe the local-projection specifications used to estimate relative and level responses, and then present the firm-level evidence.

4.1 | Empirical Strategy

Relative response across firms. We estimate relative markup responses across firm groups using panel local projections (LP – Jordà (2005)) by ordinary least squares:

$$\Delta_h \log \mu_{i,t+h} = \sum_{x \in \mathcal{X}} \left(\alpha_{x,h} + \beta_{x,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{x,h}^k \varepsilon_{t+k}^m \right) \times \mathbf{1}_{i \in I^x} + \sum_{\ell=1}^L \delta'_h X_{i,t-\ell} + \varphi_{i,h} + \varphi_{s,t,h} + \vartheta_h t + u_{i,t+h}, \quad (5)$$

with horizons given by $h = 0, 1, \dots, H$. The dependent variable is the cumulative change in markups for firm i at horizon h , defined by the following expression:

$$\Delta_h \log \mu_{i,t+h} \equiv \log \mu_{i,t+h} - \log \mu_{i,t-1}. \quad (6)$$

As for the independent variables, the indicator $\mathbf{1}_{i \in I^x}$ in Equation (5) equals one if firm i was above the median value of characteristic x in the previous year. The set of characteristics is

$$\mathcal{X} = \{\text{age}, \text{assets}, \text{salesshare}, \text{leverage}, \text{liquidity}, \text{markup}\}.$$

These firm-level characteristics serve as proxies for leading theories of performance heterogeneity, as discussed in Section 2. The coefficients of interest are $\gamma_{x,h}^0$, which measure the relative markup response of firms above the median of characteristic x to the contemporaneous MP shock ε_t^m .

Our baseline specification is semi-parametric, using median dummies rather than linear interactions between firm characteristics and MP shocks, though we confirm results are similar when using a parametric specification with linear interaction terms. We include past and future MP shocks, with $\kappa = 4$, to control for serial correlation in the shock series and reduce attenuation bias in panel LPs (Teulings and Zubanov 2014). Firm fixed effects (FEs), $\varphi_{i,h}$, absorb permanent firm heterogeneity, while sector-time FEs, $\varphi_{s,t,h}$, control for aggregate and sectoral shocks, including

sectoral differences in nominal rigidities. We show that removing both sets of FEs, thereby fully exploiting permanent cross-firm variation and time variation, yields similar results. Our baseline also includes linear and quadratic trends, $\vartheta_h t$, to account for the growth in markups documented by De Loecker et al. (2020).

Following Ottonello and Winberry (2020), we interact $\mathbf{1}_{i \in I^x}$ with previous-quarter GDP growth ΔY_{t-1} to account for varying sensitivities of firm markups to the business cycle. We include this control in line with best practices, but robustness checks confirm that its inclusion does not affect the results. Additionally, the vector $X_{i,t}$ comprises: (i) firm-level controls such as sales growth and overhead costs relative to sales, (ii) macro-level controls including GDP and CPI growth, changes in the 1-year Treasury rate, and the EBP,⁵ and (iii) their lags (i.e., $L = 4$). The vector also includes fiscal-quarter dummies to address seasonality in accounting practices, as noted by Ottonello and Winberry (2020). Finally, standard errors $u_{i,t}$ are clustered at both the firm and quarter level to control for any serial correlation in the error term.⁶

Level Response Across Firms. Equation (5) identifies *relative* responses to MP shocks across firms but, because of sector-time fixed effects, does not identify the *level* response of each group. To determine whether markups are procyclical or countercyclical within each group, we hence drop sector-time FEs and estimate:

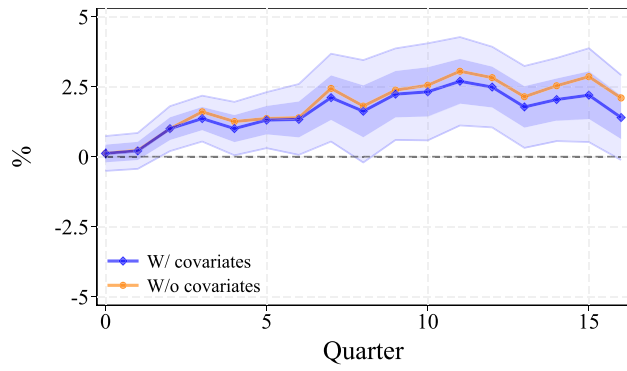
$$\Delta_h \log \mu_{i,t+h} = \alpha_h + \beta_h \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{x,h}^k \varepsilon_{t+k}^m + \left(\alpha_{x,h} + \beta_{x,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{x,h}^k \varepsilon_{t+k}^m \right) \times \mathbf{1}_{i \in I^x} + \sum_{\ell=1}^L \delta'_h X_{i,t-\ell} + \varphi_{i,h} + \vartheta_h t + u_{i,t+h}, \quad (7)$$

again for $h = 0, 1, \dots, H$ and $x \in \mathcal{X}$. This second specification exploits time-variation to estimate the level effect of MP shocks on markups, as captured by γ_h^0 , as well as the relative additional effect of belonging to group $x \in \mathcal{X}$, which is captured by $\gamma_{x,h}^0$. All firm and macro controls, trends, shock terms, and clustering are as in Equation (5).

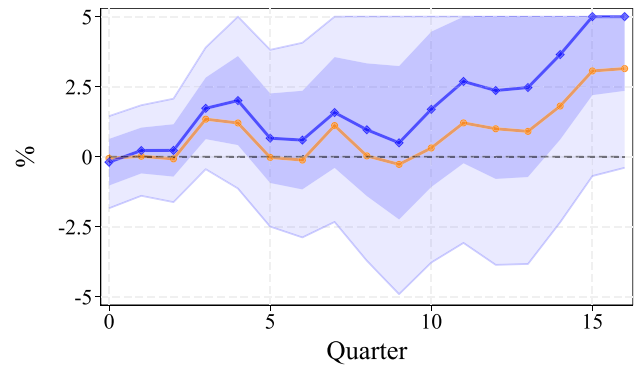
4.2 | Results

4.2.1 | Main Results

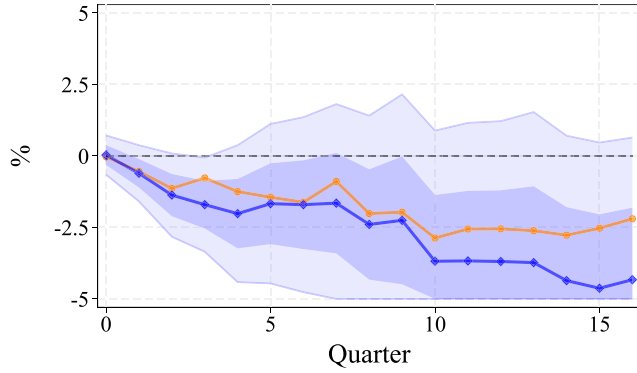
4.2.1.1 | Relative Response Across Firms. Figure 1 reports the estimated $\gamma_{x,h}^0$ coefficients from Equation (5) and horizons $h = 1, \dots, 16$, normalized to a 25-basis-point contractionary MP shock, along with confidence intervals around point estimates. Firm age is the clearest predictor of heterogeneous markup responses: firms above the median age increase their markups relative to younger firms, with the difference reaching 2.69 percent and becoming statistically significant. Sales share and size (i.e., assets) also display suggestive differences, but these are smaller and only weakly significant. By contrast, leverage, liquidity, and the level of markups do not robustly predict heterogeneous



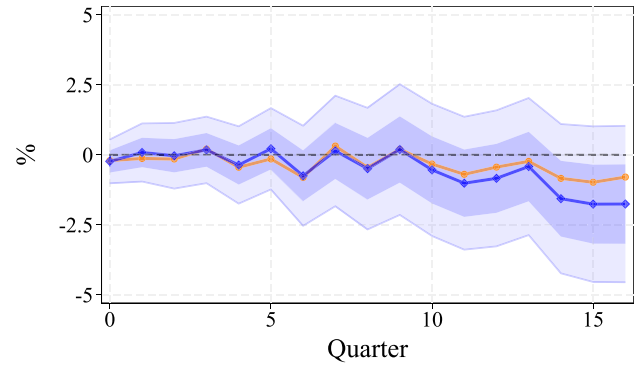
(a) Age



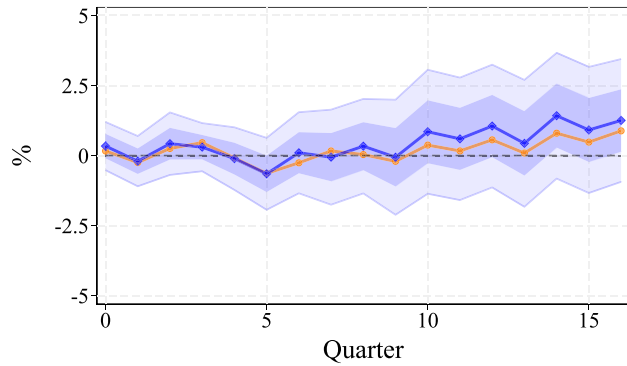
(b) Sales Share



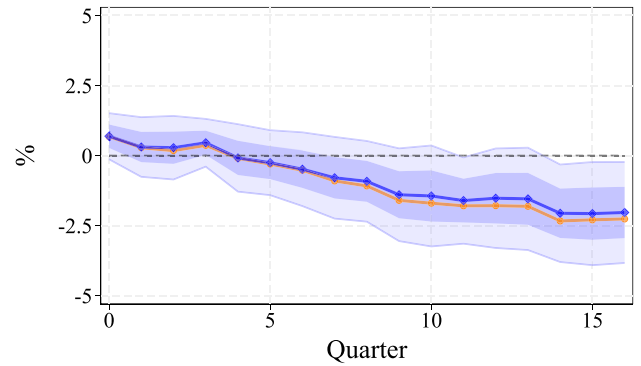
(c) Assets



(d) Liquidity



(e) Leverage

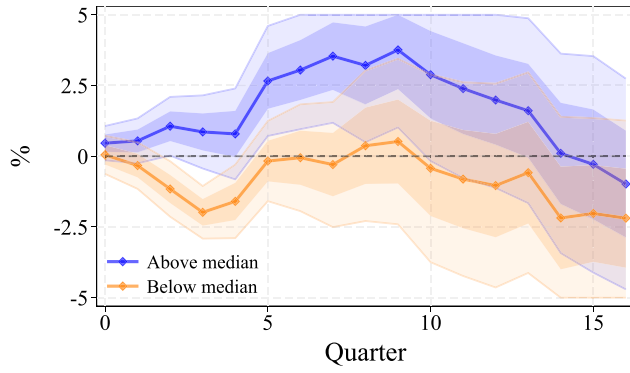


(f) Markups

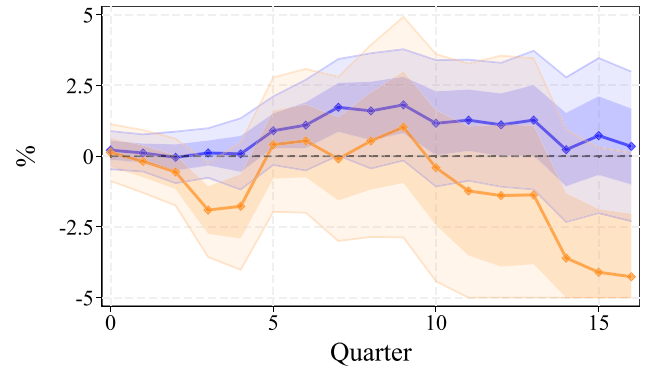
FIGURE 1 | Heterogeneous markups cyclicity by firm-level characteristics. *Notes:* Figure 1 shows the relative markup response of (a) old versus young, (b) high versus low sales share, (c) big versus small, (d) high versus low liquidity, (e) high versus low leverage, (f) high versus low markup firms conditional on a monetary policy (MP) shock. Coefficients $\hat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with and without covariates — the blue and orange lines — and for horizons $h = 1, \dots, 16$. Note that Figure 1 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68% and the 90% confidence intervals around $\hat{\gamma}_{x,h}^0$ when estimating Equation (5) with all covariates. Standard errors clustered at the firm and quarter level.

responses. The estimates are similar whether characteristics are included jointly or one at a time — the solid blue and orange lines respectively. Online Appendix A.2.12 shows that, among the same characteristics, age is also the strongest predictor of heterogeneous responses in sales and cost of goods sold, the two components of our markup measure.

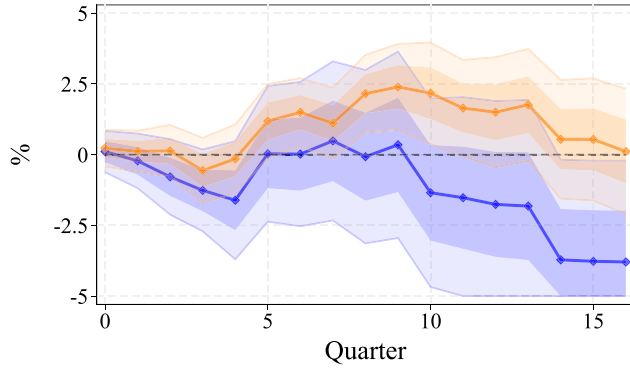
4.2.1.2 | Level Response Across Firms. Figure 2 reports the level responses estimated from Equation (7). Old firms increase markups by up to 3.76% after a 25-basis-point contractionary MP shock, while young firms reduce markups by up to 2%. Thus, old firms display countercyclical markups, while young



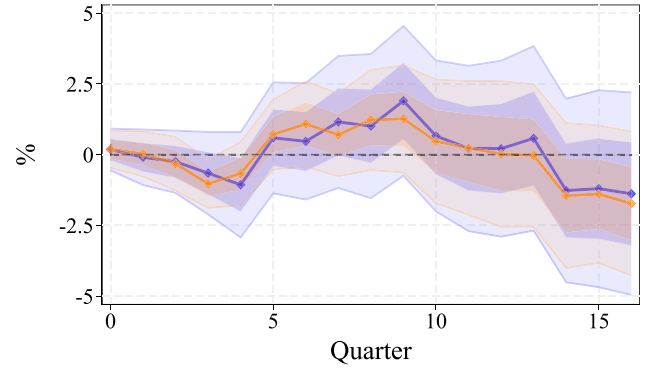
(a) Age



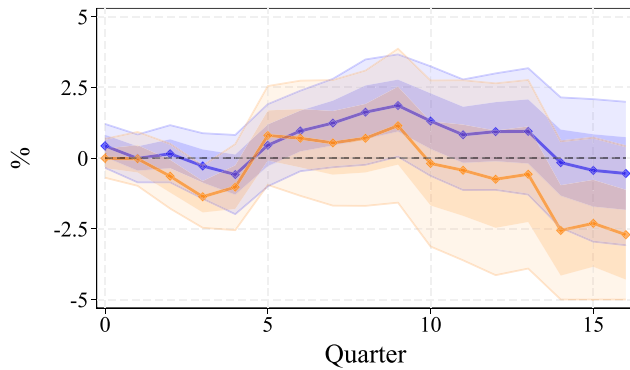
(b) Sales Share



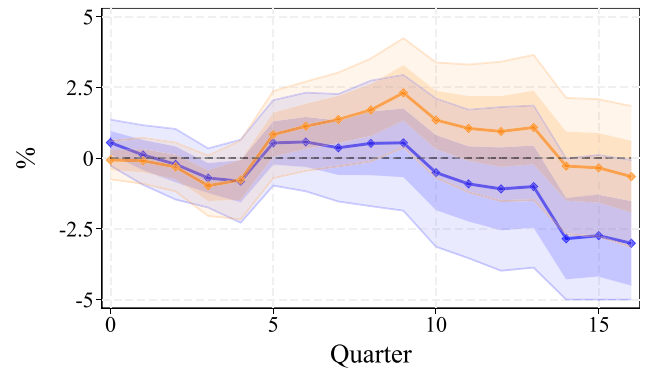
(c) Assets



(d) Liquidity



(e) Leverage



(f) Markups

FIGURE 2 | Heterogeneous markups cyclicalities across firms. *Notes:* Figure 2 shows the markup response of (a) old and young, (b) high and low sales share, (c) big and small, (d) high and low liquidity, (e) high and low leverage, (f) high and low markup firms conditional on a monetary policy (MP) shock. We plot the response of firms above (blue) and below (orange) the median in each category. Coefficients $\hat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Note that Figure 2 is normalized to a 25-basis-point contractionary MP shock. The dark and light-shaded areas report the 68% and the 90% confidence intervals. Standard errors clustered at the firm and quarter level.

ones display procyclical markups. Firms with larger sales shares show a similar pattern, but the estimates are smaller and less precise. Panel (c) of Figure 2 highlight instead that firms below median assets show a more countercyclical markup response to MP shocks, with markups of low-asset firms significantly increasing between q5 and q10 after a 25-basis-point monetary tightening. Moreover, firms below the median markup seem to adjust downwards their markups in the short run and increase in

the medium run, though coefficients are statistically significant only in a couple of quarters. Firms below the median leverage and liquidity have a qualitatively similar response in the short run, but none of the estimated coefficients is significant in the medium run.

4.2.1.3 | Summary of Findings. We show that firm age strongly predicts markup cyclicalities conditional on MP shocks:

markups of old firms are countercyclical, while markups of young firms are procyclical. Once age is accounted for, other characteristics — assets, sales share, leverage, liquidity, and markups — have more limited predictive power.

4.2.2 | Robustness Analysis

Online Appendix A.2 reports a battery of robustness checks for Equation (5); analogous checks for Equation (7) are available upon request. We exclude the ZLB period, omit future MP shocks, use linear interactions instead of median dummies, employ the MP shock series from Gürkaynak et al. (2005), measure age using Jay Ritter's founding-year data, define firm groups within sector-quarter cells, compute markups using Cobb-Douglas and Translog elasticities, use Lerner-index markups with constant and non-constant returns to scale, remove controls, remove all fixed effects, estimate elasticities using dynamic-panel methods by Arellano and Bond (1991) and Blundell and Bond (1998) to compute alternative markup measures, and exclude firms in the financial sectors from our estimation sample. Across these exercises, age remains a robust predictor of heterogeneous markup responses to MP shocks, although some alternative markup measures attenuate statistical significance.

4.2.3 | Results Interpretation and Discussion

Our goal is not to microfound why age predicts heterogeneous markup responses to MP shocks, but to identify the firm characteristics that matter empirically and trace their aggregate implications through changes in the firm distribution. Still, the age gradient is consistent with two mechanisms discussed in Section 2. First, age may proxy for financial frictions. Young firms are more likely to lack stable cash flows, long credit histories, and unconstrained access to external finance (Haltiwanger et al. 2013; Davis and Haltiwanger 2024; Cloyne et al. 2023; Dinlersoz et al. 2018). After a contractionary MP shock, such firms may face rising marginal costs or cut prices to generate cash flows, consistent with procyclical markups and with the evidence in Kim (2021).

Second, age may proxy for demand-side frictions. Older firms are more established in their markets, have accumulated customers and intangible capital, and may face less elastic demand (Foster et al. 2008; Ravn et al. 2006; Argente and Yeh 2022). This interpretation is also consistent with models in which large or established firms pass through cost shocks less aggressively (Gopinath and Itskhoki 2010; Klenow and Willis 2016; Mongey 2021; Wang and Werning 2022; Baqaee et al. 2024; Alvarez et al. 2022).

In conclusion, our evidence suggests that both demand and financial frictions may explain the role of age in driving the differential response of firm markups to MP shocks, and more granular data would be necessary to carefully disentangle these channels. We stress again that the specific reason behind the significance of firm age for the cyclicalities of markups does not change the scope of our question, as we rather aim to highlight the contribution of the documented heterogeneity in

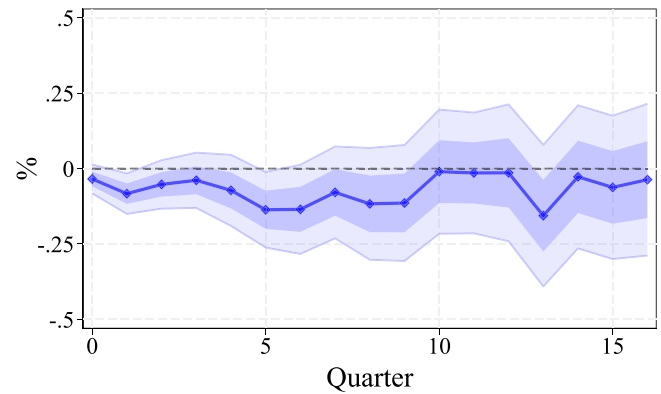


FIGURE 3 | Heterogeneous cost shares cyclicalities. *Notes:* Figure 3 shows the variable cost shares response of old firms conditional on a monetary policy (MP) shock. Coefficients $\hat{\gamma}_{x,h}^0$ are shown for the estimation of Equation (5) with covariates and for horizons $h = 1, \dots, 16$. Figure 3 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68% and the 90% confidence intervals around $\hat{\gamma}_{x,h}^0$. Standard errors clustered at the firm and quarter level.

firms' markups cyclicalities to the cyclicalities of the aggregate U.S. markup conditional on MP shocks.

4.2.4 | Reallocation Across Firm-Age Groups

Having established firm age as a plausible predictor of heterogeneous firm-level responses of markups to MP shocks, we go back to the conceptual framework outlined in Section 2 and examine the reallocation component highlighted by Equation (3). We re-estimate Equation (5) using firm-level variable-cost shares, rather than markups, as the dependent variable. Since variable-cost shares sum to one across age groups by construction, the relative response of old firms is also their level cost-share response. Figure 3 presents the resulting response.

Figure 3 shows that the variable-cost share of older firms falls by about 0.1 percent after a 25-basis-point contractionary MP shock, implying a corresponding increase for younger firms. The estimate is only weakly significant, but it points to a modest reallocation of economic activity from old to young firms, consistent with the reallocation from large to small firms in Baqaee et al. (2024). Online Appendix A.3.2 shows that the same pattern emerges when using sales shares instead of variable-cost shares.

The next section combines the micro-level evidence on the heterogeneity in the conditional cyclicalities of markups by firm age with evidence on the aging of U.S. firms to derive potential implications for the conditional cyclicalities of the aggregate markup.

5 | Aggregate Implications

In this section, we first explain how we aggregate the previously discussed firm-level markup responses to gauge at the cyclicalities of the aggregate markup in response to MP shocks across our entire sample. Next, we explore how shifts in the distribution of firms towards older ones could shape the cyclicalities of the

aggregate markup conditional on MP shocks in different periods of our sample. We then conclude by examining some additional aggregate implications carried out by our findings.

5.1 | Aggregate Markup Cyclicity

We begin by arguing that the firm-level analysis in Section 4 is a starting point to explore the cyclicity of the aggregate markup. Given that one dimension of heterogeneity - for example firm age - proved potentially significant in predicting the response of markups to MP shocks, we redefine the aggregate markup from Section 2 as:

$$\mathcal{M} = \omega_O^v \mu_O + (1 - \omega_O^v) \mu_Y, \quad (8)$$

where ω_O^v is the variable-cost share of old firms, defined as firms above the median age, while μ_O and μ_Y are the variable-cost-weighted markups of old and young firms. Log-linearizing Equation (8) around the steady state conditional on a MP shock gives:

$$\begin{aligned} \frac{\partial \Delta_h \log \mathcal{M}_h}{\partial \varepsilon^m} &= \underbrace{\frac{\omega_O^v \mu_O}{\mathcal{M}} \frac{\partial \Delta_h \log \mu_{O,h}}{\partial \varepsilon^m} + \frac{(1 - \omega_O^v) \mu_Y}{\mathcal{M}} \frac{\partial \Delta_h \log \mu_{Y,h}}{\partial \varepsilon^m}}_{\text{Direct effect}} \\ &+ \underbrace{\frac{\mu_O - \mu_Y}{\mathcal{M}} \frac{\partial \Delta_h \omega_{O,h}^v}{\partial \varepsilon^m}}_{\text{Indirect effect}} \end{aligned} \quad (9)$$

where $\partial \Delta_h \log \mu_{O,h} / \partial \varepsilon^m$, $\partial \Delta_h \log \mu_{Y,h} / \partial \varepsilon^m$, and $\partial \Delta_h \omega_{O,h}^v / \partial \varepsilon^m$ represent the IRFs of markups of old and young firms, and the old firms' variable cost share, respectively, as estimated in Section 4 conditional on MP shocks. Equation (9) is indeed the two-group counterpart of Equation (3): the direct effect captures heterogeneous markup responses across age groups, while the indirect effect captures reallocation in variable-cost shares.

Using our baseline age measure, $\omega_O^v = 0.76$, $\mu_O = 1.30$, $\mu_Y = 1.21$, and $\mathcal{M} = 1.28$. Using Jay Ritter's founding-year age measure, the corresponding values are 0.66, 1.33, 1.24, and 1.29. Combining these moments with the impulse responses estimated in Section 4 and Equation (9), we compute the aggregate markup response to MP shocks, as shown in Figure 4.

Figure 4 shows that the aggregate markup rises after a 25-basis-point contractionary MP shock. The response averages about 3% between q5 and q10 and is significant at the 90% level, consistent with the countercyclical markup response predicted by sticky-price NK models and with the evidence in Bils et al. (2018). The response is similar when using Jay Ritter's age measure. Moreover, Online Appendix A.3.1 shows that the IRF in Figure 4 is primarily driven by the direct effect, meaning that the heterogeneous impact across different firm-age groups is what matters for the conditional cyclicity of the aggregate markup. The indirect effect, while determining the reallocation of costs across firms discussed in Section 4.2.4, is quantitatively small because old and young firms have similar average markups, so $(\mu_O - \mu_Y) / \mathcal{M}$ is close to zero.

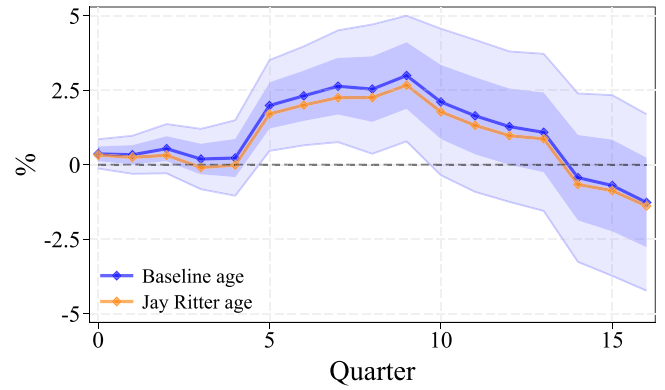


FIGURE 4 | Aggregate markup cyclicity. *Notes:* Figure 4 presents the aggregate markup response conditional on a monetary policy (MP) shock, derived using Equation (9). The results in Figure 4 are normalized to a 25-basis-point contractionary MP shock. The dark blue line represents the results obtained using our baseline measure of age, while the orange line represents the results using the Jay Ritter age measure. The dark and light blue shaded areas indicate the 68% and 90% confidence intervals around the point estimates, respectively. A delta method based on Equation (9) is used to construct confidence intervals around the point estimates.

Finally, although the most robust weighting method for the aggregate markup involves variable costs (e.g., Grassi 2017; Edmond et al. 2023), Online Appendix A.3.2 demonstrates that using sales weights to compute the aggregate markup, as proposed by De Loecker et al. (2020), yields similar results. This holds true for (i) estimates of the reallocation of economic activity across firm-age groups after MP shocks, (ii) the aggregate markup's IRF conditional on the same shocks, and (iii) estimates of the strength of the direct and indirect effects for the cyclicity of the aggregate markup.

Since we have argued for the importance of the firm distribution for the response of the aggregate markup, as per the explicit aggregation in Equation (9), we next investigate how changes in the age distribution of firms may influence the conditional cyclicity of the aggregate markup, relating to existing findings in the literature.

5.2 | Changing Markups Cyclicity: Firm Aging

5.2.1 | Micro-to-Macro Approach

Equation (9) implies that aggregate markup cyclicity depends not only on firm-level markup responses and reallocation, but also on the relative size of each firm-age group. This is important because a growing literature documents a decline in firm formation and in the exit rate of older firms, leading to an aging of the U.S. firm population (e.g., Hathaway and Litan 2014; Decker et al. 2016). While Hopenhayn et al. (2022) emphasize the implications of firm aging for aggregate factor shares, our framework suggests that firm aging may also affect business-cycle dynamics through markup cyclicity. Table 1 summarizes the underlying age-distribution moments.

TABLE 1 | Firm-age distribution and aggregate markups

	1990-2000	2001-2016
Using years since incorporation		
Average firm age	6	13
Share of old firms	20%	56%
Cost share by old firms (ω_o)	60%	82%
Markup old firms (μ_o)	1.30	1.30
Markup young firms (μ_y)	1.22	1.20
Aggregate markup ($\mathcal{M}$)	1.27	1.28
Using Foundation Year by Jay Ritter		
Average firm age	20	27
Share of old firms	26%	54%
Cost share by old firms (ω_o^v)	48%	76%
Markup old firms (μ_o)	1.33	1.33
Markup young firms (μ_y)	1.23	1.25
Aggregate markup ($\mathcal{M}$)	1.27	1.31

Note: Data is from Compustat 1990-2016. Cost shares are computed using the variable input measure employed in the empirical analysis, that is, cost of goods sold. The markup for young and old firms is the average variable cost-weighted markup for firms above and below the median age, respectively. The aggregate markup is calculated as the sum of the markups for old and young firms, each weighted by their respective variable cost shares.

Table 1 confirms that firms in our sample age substantially over time. Between 1990–2000 and 2001–2016,⁷ the share of old firms roughly doubles under both age measures, and their variable-cost share rises from 60% to 82% using Compustat age, and from 48% to 76% using Jay Ritter’s founding-year measure. Average markups are stable across subperiods and align with the evidence in De Loecker et al. (2020), so the key change for aggregation is the growing weight of old firms. Figure 5 presents the implied aggregate responses before and after 2000.

To uncover the implications of firm aging on the conditional cyclicity of the aggregate markup, we then follow the procedure outlined in Section 5.1 using the data from Table 1, and recompute Equation (9) separately before and after 2000. Figure 5 shows that the aggregate markup is mostly acyclical before 2000, with point estimates leaning mildly procyclical. After 2000, it becomes mildly countercyclical under both age measures, with positive and statistically significant responses from q5 to q10. Online Appendix A.3.3 further shows that the differential markup response of old relative to young firms does not materially change across the two subperiods. Thus, the shift in aggregate markup cyclicity should be driven mainly by changes in the firm age distribution, rather than by changes in the responsiveness of a given firm-age group. Through the lens of the NK model, this could indicate a channel through which amplification forces are intensifying in the U.S. economy. Next, we validate this finding by further employing a direct aggregate approach to assess the aggregate markup cyclicity.

5.2.2 | Macro Validation of the Micro-to-Macro Approach

We validate the micro-to-macro evidence by computing the response of the aggregate markup directly. Specifically, we estimate the following aggregate LP:

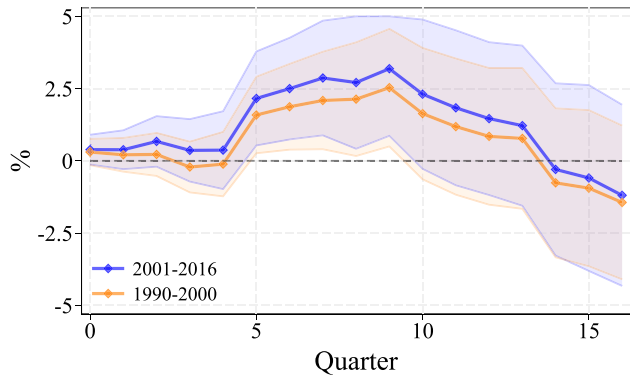
$$\Delta_h \log \mathcal{M}_{t+h} = \alpha_h + \left(\alpha_{t>2000q4,h} + \beta_{t>2000q4,h} \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_{t>2000q4,h}^k \varepsilon_{t+k}^m \right) \times \mathbf{1}_{t>2000q4} + \beta_h \Delta Y_{t-1} + \sum_{k=-\kappa}^h \gamma_h^k \varepsilon_{t+k}^m + \sum_{\ell=1}^L \delta_h' X_{t-\ell} + \vartheta_h t + u_{t+h}, \quad (10)$$

with horizons given by $h = 0, 1, \dots, 16$. The dependent variable is the cumulative change in the aggregate markup, namely the cost-weighted average of firms’ markups. The interaction between MP shocks and the dummy $\mathbf{1}_{t>2000q4}$ captures the change in aggregate markup cyclicity after 2000. Controls, lags, and trends follow the firm-level specifications from Section 4, with additional dummies for the Great Financial Crisis and the ZLB. Standard errors are Newey–West with three lags,⁸ to control for autocorrelation. For robustness, we also interact the MP shocks with a linear trend instead of the post-2000q4 dummy to capture changes in the response of $\mathcal{M}$ over time. Figure 6 reports these estimates.

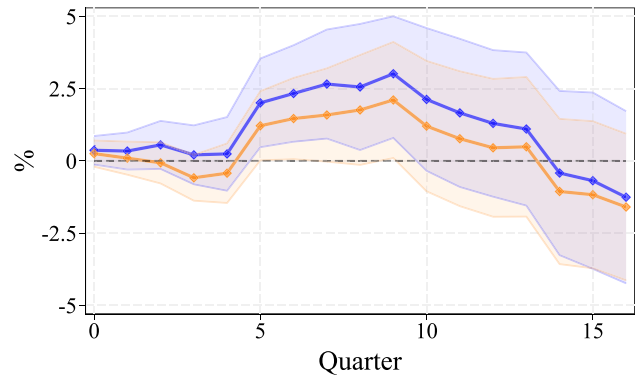
Figure 6 reports the aggregate markup response to a 25-basis-point contractionary MP shock before (Figure 6(a)) and after 2000 (Figure 6(b)), plus the estimated difference between the two IRFs (Figure 6(c)). Results echoes the micro-to-macro findings in Figure 5: the aggregate markup response shifts from mildly procyclical before 2000 to mildly countercyclical afterward, and the difference between the two IRFs is significant at the 90% level until q4. To better compare the two approaches, we also compute the cumulative impulse response function (CIR), which measures the area under the IRF and summarizes the impact and persistence of the economy’s response (Alvarez et al. (2016)). Under the micro-to-macro approach, the CIR of the pre/post difference in Figure 5 is 8.41 using Compustat age, 11.62 using Jay Ritter’s age-measure. Under the aggregate-economics approach, the CIR for Panel (c) in Figure 6 is 7.19.⁹

5.2.3 | Changing Markups Cyclicity in Relation to the Literature

The results in Sections 5.2.1 and 5.2.2 suggest that the period of analysis matters when assessing the conditional cyclicity of the aggregate markup. Changes in the age structure of the U.S. firms began in the 1980s, so the growing role of older firms – whose markups are countercyclical to MP shocks – may have become increasingly relevant for aggregate dynamics. To highlight the role of sample selection, rather than provide an exhaustive review, we compare our findings with two influential aggregate-data studies: Bils et al. (2018), which argues for countercyclical markups and aligns with our full-sample results in Section 5.1, and Nekarda and Ramey (2020), which finds markups to be mildly procyclical or acyclical in response to MP shocks.

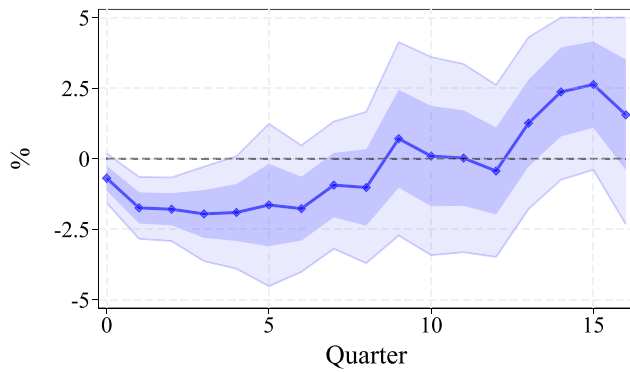


(a) Compustat Age

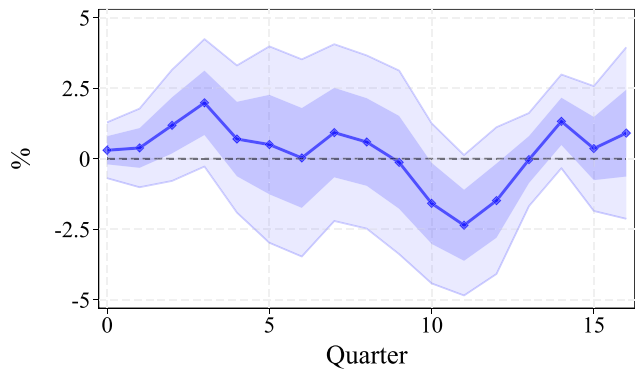


(b) Jay Ritter Age

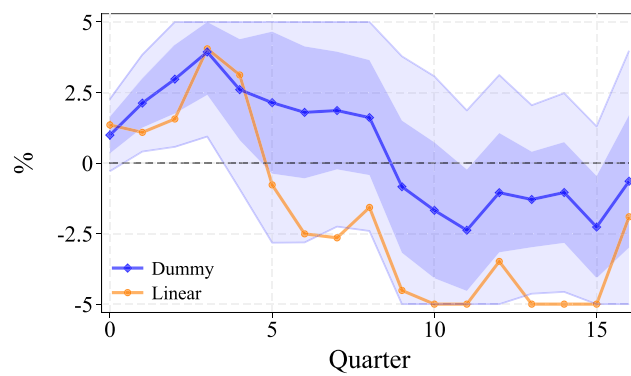
FIGURE 5 | Firm aging and aggregate markup cyclicity – Micro-to-Macro Approach. *Notes:* Figure 5 shows the markup response to a monetary policy (MP) shock before and after 2000 for changes in the age distribution of firms using (a) Compustat age and (b) Jay Ritter age. Figure 5 is normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68 percent and the 90 percent confidence intervals around $\widehat{\gamma}_{x,h}^0$ estimating Equation (5) with all covariates. Standard errors clustered at the firm and quarter level. A delta method based on Equation (9) is used to construct confidence intervals around point estimates.



(a) Before 2000



(b) After 2000



(c) Difference

FIGURE 6 | Aggregate markup cyclicity macro approach. *Notes:* Figure 6 presents the aggregate markup response to a contractionary monetary policy (MP) shocks before (Figure 6(a)) and after 2000 (Figure 6(b)) and the estimated difference (using a dummy or a linear specification) between the two impulse responses (Figure 6(c)). Impulse responses are normalized to a 25-basis-point contractionary MP shock. The dark and light blue areas report the 68% and the 90% confidence intervals around $\widehat{\gamma}_{x,h}^0$. Standard errors are calculated following Newey-West with 3 lags.

A key difference between these papers is the sample period. [Bils et al. \(2018\)](#) focus on a sample starting in the late 1980s, whereas [Nekarda and Ramey \(2020\)](#) use data going back to the 1950s.¹⁰ As discussed above, these periods feature different firm distributions: the former is increasingly dominated by older firms, whose markups are countercyclical, while the latter places more weight on younger firms, whose markups lean procyclical. Differences in sample periods may therefore reconcile these divergent findings, and support our micro-to-macro approach, which estimates the primitive firm-level markup responses to MP shocks and highlights the role of aggregation in shaping aggregate markup cyclicalities.

5.3 | Magnitudes of the Findings and Macro Implications

We close with an admittedly partial discussion of two potential implications of our methodology for macroeconomic frameworks: monetary policy transmission and the relative importance of demand and supply shocks in driving business cycles.

5.3.1 | Monetary Policy Transmission

In the NKPC, inflation depends on current and expected future markup gaps. Hence, the same firm-level markup responses that shape aggregate markup cyclicalities may also inform the inflation response to MP shocks. Following common approaches to aggregate price indexes (e.g., Divisia price index and PPI) and [Baqae et al. \(2024\)](#), define aggregate inflation as the sales-weighted average of firm-level inflation:

$$\pi_t = \sum_i \omega_{i,t}^s \pi_{i,t}, \quad (11)$$

where $\omega_{i,t}^s$ are sales weights. Combining Equation (11) with the firm-level NKPC in Equation (1), and splitting firms into old and young groups, gives:

$$\begin{aligned} \frac{\partial \pi_t}{\partial \varepsilon_t^m} &= \omega_o^s \frac{\partial \pi_{o,t}}{\partial \varepsilon_t^m} + (1 - \omega_o^s) \frac{\partial \pi_{y,t}}{\partial \varepsilon_t^m} \\ &= -\omega_o^s \lambda_o \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t \left[\frac{\partial \Delta_h \log \mu_{o,t+h}}{\partial \varepsilon_t^m} \right] \\ &\quad - (1 - \omega_o^s) \lambda_y \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t \left[\frac{\partial \Delta_h \log \mu_{y,t+h}}{\partial \varepsilon_t^m} \right] \end{aligned} \quad (12)$$

Equation (12) provides a simple comparative static linking the estimated markup responses, the sales share of old firms, and the inflation response to MP shocks. We implement it using the estimated markup IRFs of old and young firms from Equation (7), the sales shares of old firms, a quarterly discount factor $\beta = 0.99$, and a common NKPC slope $\lambda_o = \lambda_y = 0.006$ from [Hazell et al. \(2022\)](#). Table 2 reports the resulting values.

Table 2 reports the results of this exercise. Across age definitions, the implied aggregate inflation response to a 25-basis-point contractionary MP shock ranges from -0.07 to -0.10 percentage points over the full sample, within the range of previous estimates

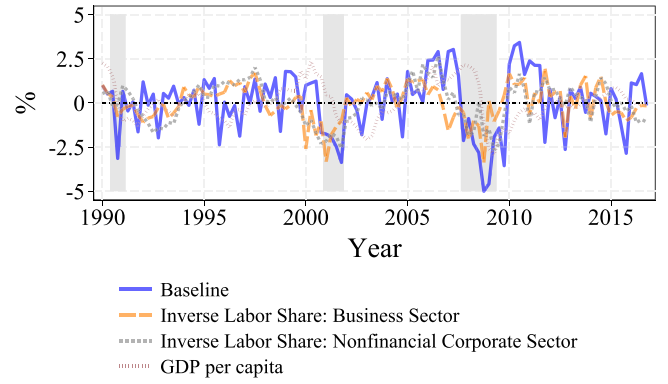


FIGURE 7 | HP-filtered aggregate markup measures. *Notes:* Figure 7 shows (i) GDP per capita, (ii) the aggregate markup derived from the estimates of firm-level markups in Compustat between 1990 and 2016, and (iii) alternative HP-filtered measures of the aggregate markup based on the inverse of the labor share, each in percentage deviation from its trend.

(e.g., [Galí and Gertler 1999](#); [Christiano et al. 2005](#); [Nakamura and Steinsson 2018](#)). The response is stronger after 2000, because old firms account for a larger sales share (between 65 and 76 percent — similar to their cost share in Table 1) and have countercyclical markups. Thus, changes in aggregate markup cyclicalities may be quantitatively relevant for monetary transmission through the lens of standard NK models.¹¹

This exercise is only a comparative static. In particular, it holds the firm-level NKPC slope λ_i fixed, whereas firm aging could itself affect price sensitivity if λ_i is an equilibrium object rather than a primitive (e.g., [Baqae et al. 2024](#)). For instance, if older firms face more inelastic demand or lower cost pass-through, their growing weight could flatten the effective NKPC and offset the mechanism illustrated above. Viewed in this light, this section should be interpreted as a proof of concept aimed at assessing the quantitative relevance of changes in aggregate markup cyclicalities, holding all else equal. A full empirical and structural assessment of how monetary policy transmission has evolved over time, allowing firm behaviors and equilibrium objects to adjust endogenously, lies beyond the scope of the present analysis.

5.3.2 | Demand Versus Supply Shocks

We conclude with a brief note on how the unconditional cyclicalities of the aggregate markup, together with its conditional cyclicalities to demand shocks studied in our empirical analysis, sheds light on which shocks are likely to drive the business cycle. Figure 7 compares the relevant aggregate series.

To this end, Figure 7 plots GDP per capita against our baseline aggregate markup and alternative inverse-labor-share measures, all expressed as percentage deviations from HP-filtered trends. These measures are unconditionally procyclical. By contrast, our estimates show that aggregate markups are countercyclical conditional on demand, or MP, shocks. If demand shocks were the only drivers of business cycles, aggregate markups should therefore be unconditionally countercyclical. Their unconditional procyclicalities instead to an important role for supply shocks, which

TABLE 2 | Firm aging and the inflation response to monetary policy shocks

	1990-2016	1990-2000	2001-2016
<i>Using years since incorporation</i>			
Sales share by old firms (ω_o^s)	76%	62%	82%
Aggregate inflation response to MP shocks ($\partial\pi_t/\partial\varepsilon_t^m$)	−0.10 p.p.	−0.07 p.p.	−0.11 p.p.
<i>Using Foundation Year by Jay Ritter</i>			
Sales share by old firms (ω_o^s)	65%	48%	76%
Aggregate inflation response to MP shocks ($\partial\pi_t/\partial\varepsilon_t^m$)	−0.07 p.p.	−0.03 p.p.	−0.10 p.p.

Note: Compustat data, 1990-2016. Sales shares are computed for firms above and below the median age, across two definitions. The aggregate inflation response is calculated as per Equation (12), with $\beta = 0.99$, $\lambda_i = \lambda = 0.006$ and using the estimated markup response of old firms from Equation (7).

generate procyclical markups even in standard NK models. This provides an additional benchmark for business-cycle models with firm heterogeneity.

6 | Conclusion

This paper revisits the question of the cyclicity of the aggregate markup through a micro-to-macro lens. Rather than studying unconditional co-movements, whose interpretation depends on the shocks driving the cycle, we focus on identified MP shocks – a class of demand shocks – and decompose the aggregate response into firm-level markup adjustments, reallocation of economic activity, and aggregation channels.

Using quarterly Compustat data from 1990q1 to 2016q4, we document substantial heterogeneity in conditional markup cyclicity by firm age. After a contractionary MP shock, young firms lower markups, whereas older firms raise them; economic activity also reallocates modestly from old to young firms. Aggregating these responses with variable-cost weights implies that the aggregate markup is countercyclical to MP shocks over the full sample, but this average masks an important change over time. As the U.S. firm distribution shifted toward older firms, aggregate markup cyclicity moved from acyclical or mildly procyclical in the early part of the sample to countercyclical in the later period. Direct aggregate estimates support the same pattern.

These findings help reconcile conflicting evidence in the aggregate markup literature and suggest that changes in the firm distribution can matter for business-cycle dynamics. Going forward, we view our findings as providing a set of firm-level empirical benchmarks for macroeconomic models of price setting and monetary transmission. By documenting heterogeneity in markup cyclicity across firms and showing how changes in the firm distribution shape aggregate outcomes, our results highlight dimensions of firm behavior that are typically abstracted from in standard New Keynesian frameworks. We hope our evidence will inform the development of micro-founded New Keynesian models with firm heterogeneity that replicate the observed markup dynamics, and spur further research on how changes in the firm population affect monetary policy transmission. These are exciting avenues for future studies.

Acknowledgments

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Data Availability Statement

The code that produces all results in this paper, all data that the authors are permitted to redistribute, and detailed instructions for constructing the remaining inputs, are openly available in the openICPSR repository at <https://doi.org/10.3886/E251593V1>.

The firm-level data that support the findings of this study are from Compustat (S&P Global Market Intelligence), obtained through Wharton Research Data Services (WRDS). Restrictions apply to the availability of these proprietary data, which were used under license for this study and therefore cannot be redistributed by the authors; researchers with a valid WRDS/Compustat license can reconstruct the extract using the instructions in the replication package. All other data used in the paper are publicly available: the monetary policy shock series of Jarociński and Karadi (2020) and of Gürkaynak, Sack, and Swanson (2005); the production-function output elasticities of De Loecker, Eeckhout, and Unger (2020); firm founding dates from Jay Ritter's database; and macroeconomic and labor-share series from FRED (Federal Reserve Bank of St. Louis) and the U.S. Bureau of Labor Statistics. The provenance of each input is documented in the replication package.

Endnotes

¹This is our baseline measure because our focus is on markup changes, for which De Ridder et al. (2026) show that revenue- and output-based measures are highly correlated. To address the concerns in Bond et al. (2021), we also verify our findings using the accounting markup measure of Baqaee and Farhi (2020).

²Baqaee et al. 2024 theoretically discuss the macroeconomic role of reallocation effects.

³This observation is also consistent with the granularity literature pioneered by Gabaix (2011).

- ⁴The Lerner index allows us not to rely on data on the cost of goods sold for the markup estimation, and hence on the assumption that this variable captures frictionless factors. The Lerner index uses instead information on sales and operating income before depreciation, as discussed in Online Appendix A.2.8.
- ⁵We include these aggregate controls, despite the fact that sector-time fixed effects should absorb them, only to maintain comparability with the specification in Equation (7) later on.
- ⁶Clustering at the firm level accommodates flexible error term dependence across time within each firm, while clustering by time is necessary to account for firm-level shocks correlated within a quarter, beyond the co-movement due to industry-level shocks captured by sector-quarter dummies. Without quarter-level clustering, confidence intervals would be significantly narrower.
- ⁷We use the year 2000 as the cutoff as it is roughly in the middle of the pre-ZLB period. Importantly, while Compustat is a sample of relatively big and old producers, the firm distribution that emerges from Business Dynamics Statistics data shows similar patterns of aging over the period of analysis, suggesting the mechanisms we uncovered may replicate in a more representative sample of firms.
- ⁸We chose 3 lags following the rule of thumb $T^{1/4} = ((2016 - 1990) \times 4)^{1/4} \approx 3$.
- ⁹This remains valid, though to a lesser extent, if we use the absolute deviation at the peak as the metric of comparison. In that case, the absolute deviation at the peak for the micro-to-macro approach would be 1 percentage point, while for the aggregate-economics approach, the difference would be around 4 percentage points
- ¹⁰The same holds for Cantore et al. (2021), who find a countercyclical labor share, that is, a procyclical aggregate markup, conditional on MP shocks in a sample dating back to the 1960s.
- ¹¹To further gauge the economic significance of this result, the difference between 0.07 and 0.11 percentage points (as implied by our calculations based on years since incorporation) might translate into roughly 60% more cumulative monetary tightening to achieve the same reduction in inflation. To fix ideas, offsetting a 0.3-percentage-point inflationary shock would require at most three 25-basis-point rate hikes if each hike lowers inflation by 0.11 percentage points, but closer to four or five hikes if the effect is only 0.07 percentage points.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: iere70109-supp-0001-SuppMat.pdf